

Ivan Makarenko

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Summary

Aspiring Product Manager with a background in automation and product development, known for enhancing manufacturing efficiency through robotic assembly innovations. Successfully improved team productivity by implementing project management trackers. Seeking to leverage technical expertise and process improvement skills to drive product excellence and strategic development in a management role.

EDUCATION

Columbia University, School of Professional Studies

Master of Science, Technology Management

Sep 2025 - Dec 2026

New York, NY

Arizona State University, Ira Fulton School of Engineering

Bachelor of Science, Engineering with a Concentration in Robotics

Aug 2020 - Dec 2024

Mesa, AZ

SKILLS

- **Programming & Software:** Python, C/C++, MATLAB, SQL, Git/GitHub, MPLAB, Ladder Logic, Tableau, Revit, AutoCAD, SolidWorks, Cadence
- **Embedded Systems & Automation:** Microcontrollers (8-bit, 16-bit, 32-bit), PCB Design, Sensor Integration, SPI/I2C Communication, Embedded C, Robotic Systems, Automation Engineering, RPA (Robotic Process Automation) CAD
- **CAD & Electrical Design:** Electrical Circuit Design, Schematic Capture, Mechanical Assembly Assistance, Computer-Aided Design (CAD), Design Optimization
- **Tools & Platforms:** Windows/Linux OS, Excel (Advanced), Visual Studio, VS Code, Arduino IDE, Multimeters
- **Languages:** Fluent in English, Fluent in Russian, Fluent in Ukrainian

WORK EXPERIENCE

Stantec

Automation Engineering Intern

Jun 2024 - Aug 2024

Chandler, AZ

- Created and revised electrical drawings in Revit and AutoCAD for multiple mining infrastructure projects, ensuring design accuracy and alignment with engineering standards.
- Updated complex Revit models to improve schematic clarity and reduce calculation errors across cross-functional teams.
- Developed and maintained Excel-based project management trackers, streamlining documentation and boosting team coordination efficiency.

Additive Desert Designs

New Product Development Intern

Jun 2023 - Aug 2023

Mesa, AZ

- Designed robotic assembly improvements in SolidWorks/CAD, enhancing manufacturing efficiency and precision.
- Built Excel-based model-tracking systems for truck part data, reducing design cycle time by 30% and enabling faster prototype iteration.
- Assembled two OTC DT-ARC 1000 robotic welding systems with OTC engineers, ensuring operational reliability and system integration quality.

Freelance

IT Systems Consultant

Jun 2019 - Present

Dnipro, Ukraine

- Provided technology support and consulting for individual clients and small businesses, including custom PC builds, software installations, and home/office network setup, leading to improved system efficiency and client satisfaction
- Built long-term client relationships through responsive service and problem-solving, leading to repeat engagements and referrals.

PROJECTS

Robotic Explorer for NASA Psyche Mission | [The Project Link](#)

Arizona State University

Jan 2024 - Dec 2024

Mesa, AZ

- Led electrical subsystem development for a robotic rover designed to simulate exploration of asteroid Psyche; collaborated with NASA mission mentors and interdisciplinary student teams.
- Designed and integrated complete wiring system, sensors, and control electronics; ensured electrical reliability through iterative prototyping and testing.
- Supported mechanical design and full-body assembly, helping deliver a fully functional terrain-navigating prototype under real-world engineering constraints.

Solar-Tracking Hat

Arizona State University

Sep 2022 - Dec 2022

Mesa, AZ

- Built a wearable solar-tracking device using a 32-bit microcontroller to control dual SPI motors via custom PCB.
- Integrated environmental sensors and designed the PCB using Cadence, enabling automated light-source alignment for solar optimization.